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Automating Electrical Grid Asset Inspections: Improving Insulators Health Assessment with Region-Level Annotations

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SUMMARY

The increasing reliance on electrical power is intensifying the demand for efficient maintenance strategies to improve grid reliability. As the first step in predictive maintenance, visual inspection is an accessible Non-Destructive Testing (NDT) method, increasingly scaled through UAV-based remote data collection. Automating visual assessments with Computer Vision (CV) improves efficiency and consistency over manual assessments. CV models based on supervised learning have advanced performance in generic domains. However, in specific domains such as electrical grid asset inspections, annotated data is often scarce. Current annotation procedures are typically designed with a focus on asset condition reporting, which can result in a suboptimal degree of spatial detail. This work evaluates the benefits of using region-level annotations in images of defective insulators for training state-of-the-art deep learning vision models. The experiments, conducted on real-world data, explore how increasing label detail can reduce training iterations, lower data requirements, and improve test performance. The findings highlight both the advantages and challenges of using richer annotations to support the automation of visual inspection of electrical grid assets. This enables utility companies to conduct more efficient inspections, reduce costs, and ultimately improve grid reliability.

KEYWORDS

Visual Inspection, Insulator Defects, Supervised Learning, Image Classification, Object Detection.

1 Introduction

Electrical grid increasing reliability is driven by the rising dependency on electrical power. The electrical power growing demand, combined with the integration of renewable energy sources and distributed generation, not only increases the electrical grid management complexity but also requires its expansion, intensifying the demand for efficient maintenance strategies.

1.1 Electrical Grid Landscape

The global demand for electric power is projected to double by 2050, exceeding 50,000 TWh annually, with an expected 80% increase in wind and solar energy generation [1]. This shift towards renewable and distributed energy resources necessitates extensive upgrades in Transport and Distribution (T&D) systems to enable the transition away from carbon-based fuels. In fact, over the next three decades, worldwide T&D systems length is expected to exceed 160 million km [1].

While developing efficient engineering processes for designing and constructing new infrastructure is vital to meet this growing demand, the management and maintenance of existing electrical generation and distribution systems are even more critical. The lack of reliability in these systems can have severe short-term impacts, such as unexpected blackouts.

In the European Union alone, over 50% of the grid has been in operation for more than 20 years, which is approximately half of its average lifespan [1]. Since operational failures due to wear and tear tend to increase at a growing rate over time, especially as assets approach the end of their expected service life [2], aging infrastructure poses an additional risk to grid stability and demands more frequent and targeted maintenance interventions.

Among these aging components, insulators are particularly problematic, despite representing only 5 $\sim$ 8% of installation costs, they contribute to over 50% of maintenance expenses [3]. These facts highlight the urgent need for effective maintenance strategies to ensure the reliability of aging infrastructure in this sustainable energy transition.

1.2 Efficient Maintenance Strategies

The proliferation of digital technologies has reduced the need for on-site visual inspections. Digital cameras enable high-quality data capture, while digital memory allows for affordable large-scale data collection. Robotic platforms further enhance safety and efficiency by accessing remote or hazardous locations. These developments not only support convenient remote analysis but are also essential for processing the collected data using computer vision.

As inspection campaigns scale, the volume of acquired visual data increases significantly, shifting the bottleneck from data acquisition to data interpretation. In this context, automated analysis becomes essential to fully exploit the potential of modern acquisition technologies, enabling both cost reduction and improved inspection effectiveness.

Automating visual inspections improves efficiency and consistency by addressing the limitations of manual assessments. Traditional inspections require skilled personnel, and are labor-intensive and time-consuming. Moreover, human judgment is prone to inconsistency due to fatigue and cognitive bias, which can lead to errors and variability [4].

1.3 Current Limitations

While automated systems help reduce costs and improve consistency, they come with their own expenses and challenges, such as, development and maintenance costs, integration with existing systems efforts, or service failure handling overload [5]. Additionally, achieving performance that exceeds human-level accuracy remains an ongoing challenge.

Current manual approaches focus on matching a predetermined defect class identifier—often linked to a textual description of the anomaly—to a visual representation of the asset's condition. While this image classification task can be effective for generating inspection reports and guiding maintenance decisions, it may constrain the training of deep learning models by providing only coarse, image-level supervision and introducing ambiguity in images that contain multiple defects.

Early object detection models demonstrated that introducing region-level annotations can significantly improve model performance [6]. By guiding feature extraction through spatially localized supervision, these models achieved substantial gains on popular benchmarks. Region annotations offer a more informative learning signal by explicitly encoding the location of relevant features, enabling models to focus attention, reduce ambiguity, and converge more efficiently with fewer training samples.

1.4 Proposed Approach

This paper investigates the impact of region-level supervision for training state-of-the-art deep vision models to detect defective overhead line (OHL) insulators. Specifically, we evaluate how increasing the spatial granularity of annotations affects training efficiency, data requirements, and generalization performance when using real-world inspection data.

The original dataset comprises 2,378 high-resolution images (4K and 8K) containing medium voltage power line insulators in operational environments. To preserve small defect details and better exploit available pre-trained weights, images were cropped around annotated objects and resized into 1280×1280 patches, doubling the default input resolution commonly used by pre-trained YOLO models [7]. Expert-provided region-level annotations cover two classes: broken and non-defective insulators. To ensure compatibility with the object detection architectures used in all experiments, image-level supervision is obtained by expanding region-level bounding boxes to cover the full image area. At the image level, samples with no annotated defects are labelled as non-defective, while images containing one or more defective insulators are labelled as defective, enabling a controlled comparison between image-level and region-level supervision using identical imagery and model architectures.

1.5 Key Contributions

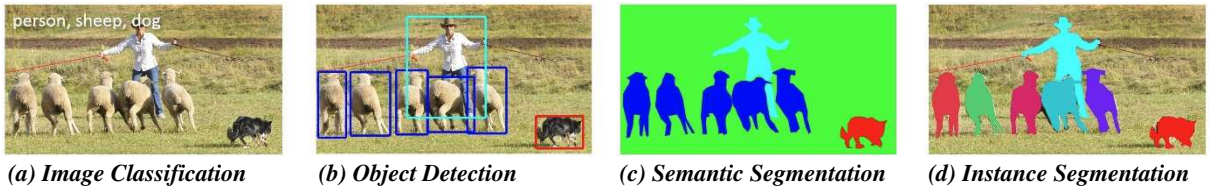
The findings demonstrate that richer, region-level annotations can significantly improve defect localization accuracy, reduce the amount of data required for training, and enable more detailed inspection reporting. At the same time, the increased annotation effort introduces practical challenges in terms of labelling cost and scalability. These results clarify the trade-offs involved in adopting spatially detailed supervision for automated inspection of electrical grid assets.

2 Related Work

In many industrial inspection systems, traditional computer vision techniques remain essential and continue to play a central role. Rule-based algorithms, geometric constraints, and handcrafted feature extraction are particularly effective in controlled environments where asset geometry, camera placement, illumination, and acquisition parameters are fixed. Under these conditions, regions of interest can be reliably identified, measurements can be explicitly constrained, and system behaviour remains predictable and interpretable. Such approaches are often preferred in production settings due to their determinism, low computational cost, ease of validation, and compliance with industrial safety and certification requirements.

Consequently, deep learning methods are frequently adopted not as a full replacement, but as a complementary component, applied to specific sub-tasks—such as anomaly classification or fine-grained defect recognition—where handcrafted rules become brittle or insufficient. In contrast, for emerging applications that involve complex patterns, subtle defects, large-scale variability, or previously unseen asset types, deep learning often becomes the primary enabling technology, opening domains where traditional computer vision was largely ineffective.

In Computer Vision (CV), visual understanding tasks are commonly organized by the granularity of supervision and output representation, as illustrated in Figure 1. Image Classification (IC) assigns a single label to an entire image and provides only global semantic information, making it suitable for tasks where the presence of a dominant object or condition is sufficient. Object Detection (OD) extends classification by localizing one or more objects through bounding boxes while simultaneously predicting their categories, enabling identification of multiple instances and their approximate spatial locations. Semantic Segmentation (SS) assigns a class label to every pixel in the image, producing dense category-level maps without distinguishing between different instances of the same class. Instance Segmentation (IS) combines detection and segmentation by predicting pixel-accurate masks for each individual object instance, providing the most detailed representation at the cost of higher annotation complexity and computational requirements.



(a) Image Classification *(b) Object Detection* *(c) Semantic Segmentation* *(d) Instance Segmentation*

***Figure 1** – Comparison of common computer vision tasks: (a) Image Classification (IC), where a single label is assigned to the entire image; (b) Object Detection (OD), which localizes and classifies objects using bounding boxes; (c) Semantic Segmentation (SS) assigns a class label to each pixel without separating instances; and (d) Instance Segmentation (IS) produces pixel-level masks for individual objects. Adapted from [8].*

2.1 Image Classification

Image-level classification emerged as the foundational task for deep learning in computer vision, largely driven by the introduction of the ImageNet [9]. The availability of millions of labelled images enabled the training of deep convolutional neural networks, most notably AlexNet [10] , which demonstrated a significant performance leap over traditional methods.

As a result, image classification became the primary benchmark for developing architectures, training strategies, and feature representations, many of which were later reused or adapted for more complex tasks such as object detection and segmentation.

Despite its success, image-level classification has inherent limitations. By assigning a single label to an entire image, IC provides only coarse supervision and cannot localize objects or regions of interest. It struggles when images contain multiple objects or defects, when relevant features occupy a small portion of the image, or when background variations introduce ambiguity. IC models also lack spatial reasoning, making them insensitive to object relationships and positions. Consequently, while effective for global image understanding, IC may fail in tasks that require fine-grained detection, localization, or instance-level differentiation, such as industrial defect inspection or multi-object scenarios.

2.2 Object Detection

Object detection extends image classification by not only predicting object categories but also localizing them with bounding boxes. It gained prominence with large-scale benchmarks such as PASCAL VOC and MS-COCO, which provided annotated images with multiple objects per image and their corresponding bounding boxes. Detection architectures such as R-CNN [6], Fast/Faster R-CNN, YOLO, and SSD emerged to address both classification and localization jointly, enabling models to handle multiple instances of various classes in complex scenes.

Despite its advantages, object detection introduces additional complexity compared to classification. It requires precise, often costly bounding-box annotations for training, and performance is sensitive to small objects, occlusions, and overlapping instances. Detection models are generally more computationally demanding and harder to train, and their accuracy can degrade when objects vary widely in scale or appear in cluttered backgrounds. Furthermore, evaluation metrics such as average precision rely on both classification and localization, making the interpretation of errors more nuanced than in image-level classification.

2.3 Literature Gap

Object detection is often more effective for reporting defects and supporting human assessment, as bounding boxes clearly highlight regions of interest. However, in scenarios where deep learning is primarily used to flag potentially defective images for human review, it remains unclear whether the additional spatial supervision from bounding-box annotations actually improves feature learning. In other words, does backpropagating through precise localization enhance the model, or could simpler image-level labels suffice for learning discriminative representations?

To the best of our knowledge, no study has systematically investigated this question. Existing works typically compare different datasets, or different model architectures, making it difficult to isolate the practical contribution of bounding-box supervision. A rigorous evaluation would require converting object detection labels to image-level labels for the same images, training both tasks on the same model, and comparing their performance using consistent metrics, an approach that, to date, appears absent from the literature.

3 Methodology

This study investigates the practical contribution of bounding-box supervision to feature learning in electrical grid asset defect inspection. Specifically, we ask: Does object detection provide additional benefits over image-level classification when the goal is to flag potentially defective images for human review? To answer this, we design a controlled experiment that isolates the effect of spatial detail on a deep model training.

As illustrated in Figure 2, the model predictions $\hat{\boldsymbol{y}}$ are influenced by the input images $\boldsymbol{x}$, the model architecture $\boldsymbol{f}$, and the target labels $\boldsymbol{y}$, which determine the loss function and the gradients used to update model parameters $\boldsymbol{\theta}$. To isolate the effect of supervision, we fix $\boldsymbol{x}$ and $\boldsymbol{f}$ across experiments, allowing any differences in learned features or performance to be attributed solely to the form of the target $\boldsymbol{y}$. Different target formulations produce distinct optimization signals, guiding the network toward different feature representations. Bounding-box annotations provide explicit spatial constraints during training, whereas image-level labels offer only global supervision. The central question is how meaningfully this localization-driven feedback alters the learned features when the final goal is image-level defect flagging.

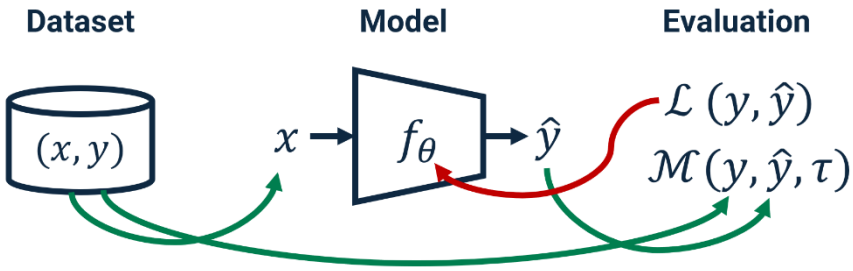


Figure 2 – Overview of supervised learning. Given a dataset of input–output pairs $(\boldsymbol{x}, \boldsymbol{y})$, a parametric model $\boldsymbol{f_\theta}$ produces predictions $\hat{\boldsymbol{y}}$ from inputs $\boldsymbol{x}$. Model parameters $\boldsymbol{\theta}$ are learned by minimizing a loss function $\mathcal{L}(\boldsymbol{y}, \hat{\boldsymbol{y}})$, whose gradient provides feedback for parameter updates during training. Model performance is evaluated using a metric $\mathcal{M}(\boldsymbol{y}, \hat{\boldsymbol{y}}, \boldsymbol{\tau})$, which may depend on a confidence threshold $\boldsymbol{\tau}$ applied to the predictions. Forward data flow is indicated by green arrows, while the backpropagation optimization feedback loop is shown in red.

3.1 Experimental Design

The experiment is designed to isolate the impact of spatial detail in the target labels while controlling for other sources of variability. Two sets of targets are considered: fitted bounding boxes tightly enclosing defects and full-image "clamped" boxes, which simulate image-level annotations. This allows us to use the same object detection model while varying only the spatial detail provided by the labels.

To quantify variability, we use 4-fold cross-validation and repeat each configuration across 3 different random seeds. These two factors capture stochastic effects and dataset variability, which we expect to introduce only noise. In contrast, the form of the target labels $\boldsymbol{y}$ is the factor expected to produce a systematic trend in the learned features and model performance. Training hyperparameters—including optimizer, learning rate, batch size, and number of epochs—are kept constant across all experiments to ensure comparability. By structuring the experiment this way, observed trends can be attributed primarily to the amount of spatial detail in the labels rather than random or dataset-induced variability.

3.2 Dataset Preparation

The dataset consists of RGB images captured in the visible spectrum during routine field inspections of electrical grid assets. Images were acquired as part of routine remote inspection workflows and subsequently assessed by domain experts to determine the condition of insulators. As a result, the dataset reflects realistic acquisition conditions, including variability in viewpoint, background, and illumination.

For the purposes of this study, the same images were re-annotated in collaboration with the inspection team to include bounding-box annotations. Domain experts manually labeled both intact (non-defective) and broken insulators, providing precise spatial localization. This additional annotation effort enabled the construction of two target formulations from an identical image set: **(i)** image-level labels indicating the presence of at least one defective insulator, and **(ii)** bounding-box labels with high degree of spatial detail.

To mitigate class imbalance, and reduce class frequency bias, additional images containing only non-defective insulators were incorporated into the dataset. These images were annotated exclusively with intact-class bounding boxes, ensuring a balanced representation of defective and non-defective samples under the full-image "clamped" label formulation.

The dataset was divided into two splits: training and testing. Both splits were further extended using supervised data augmentation. Each image was cropped to a square region defined by the minimum bounding square enclosing all annotated objects, with an additional padding factor of 0.25, to reduce excessive background content. The training split was further sorted by normalized object size and divided into four folds for cross-validation.

3.3 Metrics Definition

Standard object detection metrics, such as AP50 (Average Precision at IoU $\geq$ 50%), jointly evaluate classification and localization performance. In this context, a predicted bounding box is considered a true positive if its intersection over union with a ground-truth box meets or exceeds 50%; otherwise, it is counted as a false positive, and any ground-truth box not detected is a false negative. Precision, recall, and F1 scores can then be computed across varying confidence thresholds, producing precision–recall curves that summarize detection performance. AP50 reduces this information to a single scalar by calculating the area under the precision–recall curve, providing a widely accepted metric for object detection comparison.

However, these metrics are not fully aligned with the primary objective of this study, which is to flag potentially defective images for human review rather than to precisely localize individual defects. In particular, for models trained and evaluated with full-image "clamped" bounding boxes, localization performance is trivially high, despite offering limited practical relevance. Moreover, a model that predicts a single instance per class will always detect all defects, resulting in perfect recall regardless of actual visual pattern recognition capability.

To address this limitation, class-agnostic non-maximum suppression is applied during inference post-processing to prevent overlapping predictions from artificially inflating the score. Each trained model is then evaluated on a test set using the annotation strategy corresponding to its training (i.e., box-fit models on precise defect labels, box-all models on full-image labels).

4 Results

4.1 Dataset Description

Images were captured between 2023 and 2025 during real inspection campaigns in rural and urban distribution environments. The dataset includes ceramic and glass insulators, all annotated by domain experts as either non-defective or broken.

The original dataset of 2,378 images was first split into training and test sets to prevent data leakage. Each image was then cropped around annotated insulators and augmented, producing 3,964 training images and 792 test images while preserving the original split boundaries. The training set contains 26,573 non-defective and 3,196 broken insulator instances, while the test set contains 6,188 non-defective and 422 broken instances. Overall, the dataset comprises 4,756 images with 36,379 annotated insulator instances, of which 3,618 are broken.

Figure 3 illustrates two annotation variants were derived from the expert-provided labels: fitted bounding boxes, tightly enclosing each insulator, and full-size boxes spanning the entire image.



***Figure 3** – Representative dataset samples showing different annotation policies: (top) fitted bounding boxes for precise localization, (bottom) full-image "clamped" boxes simulating image-level supervision. Red boxes indicate broken insulators; green boxes indicate non-defective insulators.*

To further characterize the spatial properties of fitted bounding boxes, we computed the square root of each box area normalized by the corresponding square root of image area. As shown in Figure 4, the mean normalized box area is approximately 0.0779 with a standard deviation of 0.0307. Most insulators occupy between 3% and 15% of the image side length.

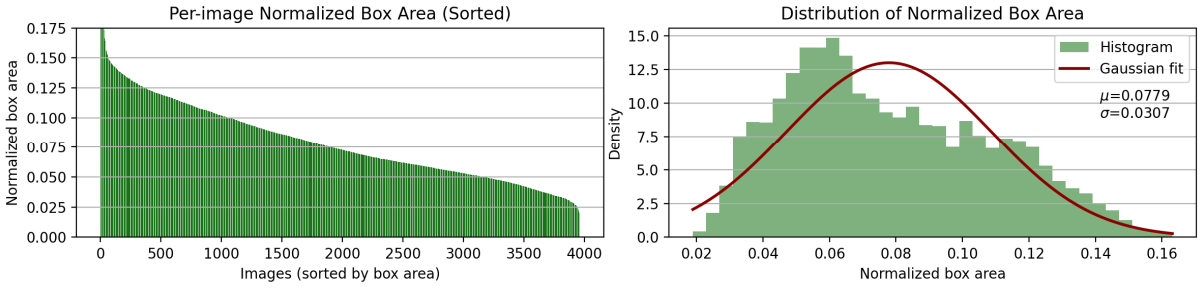


***Figure 4** – Distribution of fitted bounding-box sizes relative to image dimensions. The normalized box area for each image is sorted (left) and shown as a histogram with a Gaussian fit (right).*

4.2 Model Training

To ensure reproducibility and enable controlled comparisons between supervision types, all experiments employed the YOLOv11 medium model with consistent training hyperparameters. A detailed description of the model architecture is provided in [7]. Models were trained for 150 epochs with a batch size of 16 and input image size of 1280×1280 pixels. The first 2 layer were frozen, a dropout rate of 0.1 was applied, and a cosine learning rate schedule was used, with an initial learning rate (LR) of 0.01 and a final learning rate fraction of 0.01. A 10-epoch warmup phase was applied at the start of training. Warmup gradually increases the learning rate from a smaller value to the target LR, preventing excessively large updates in the first steps that could destabilize the network. In this setup, the relatively high initial LR during warmup also promotes early exploration of the loss landscape, helping the optimizer traverse a broader solution space and potentially avoid poor local minima.

Training stability and convergence were monitored throughout training using multiple metrics, including classification loss, AP50, precision, and recall, as shown in Figure 5. For each metric, results from all cross-validation folds are overlaid on the same axis, enabling comparisons across labelling strategies, as well trends across folds. To reduce stochastic effects, curves are averaged across repeated runs, with variability across runs illustrated by shaded regions.

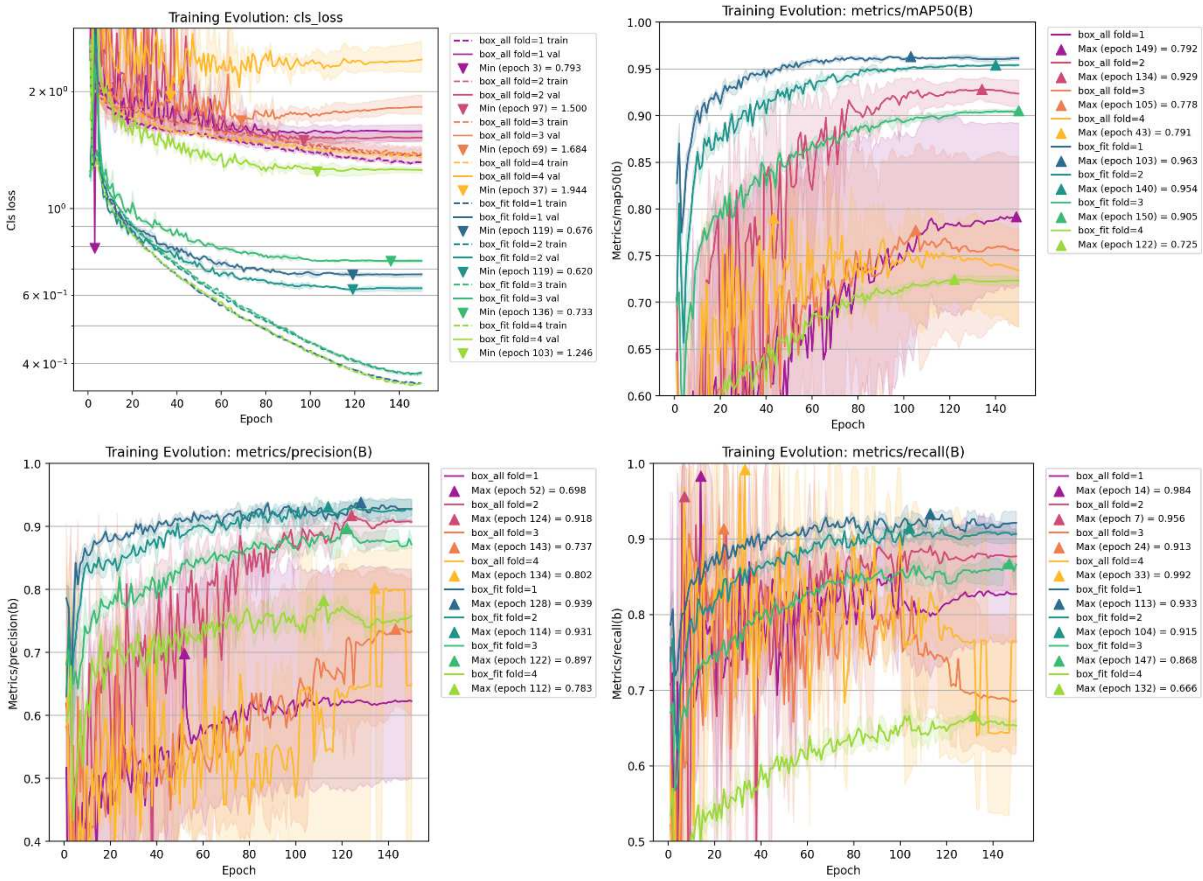


***Figure 5** - Evolution of training and validation metrics across epochs. The figure shows four subplots corresponding to classification loss, AP50, precision, and recall. For each subplot, curves from all cross-validation folds are overlaid on the same axis. Models trained with fitted bounding boxes are shown using the "viridis" colormap, while models trained with full-image clamped boxes are shown using the "plasma" colormap. For each fold, curves represent the mean across repeated runs, with shaded regions indicating the minimum and maximum values.*

4.3 Evaluation

Final model performance is evaluated on an unseen test set containing 792 images with 422 broken insulator instances. While the validation folds were ordered according to normalized bounding-box size as a proxy for viewing distance, the test set exhibits a broader and more heterogeneous distribution of object scales. Figure 6 reports F1-score, AP50, precision, and recall for each labelling protocol across cross-validation folds and repeated runs. Figure 7 presents representative prediction examples from the best-performing configuration enabling a qualitative assessment of model behaviour under realistic deployment conditions.

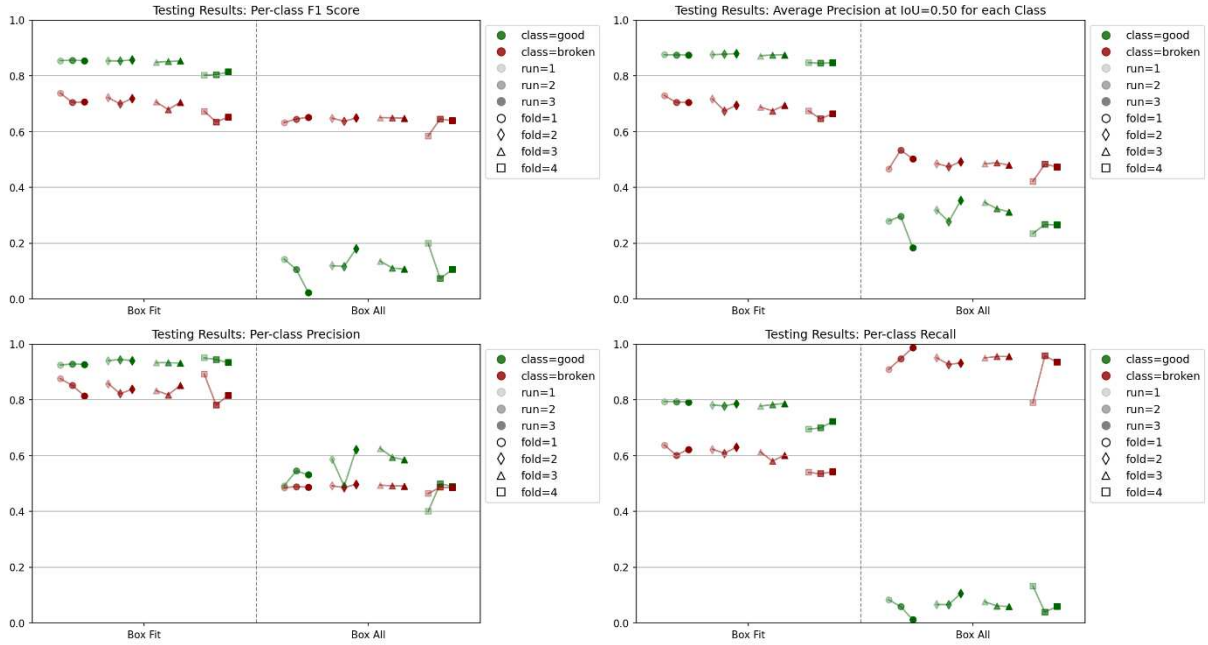


Figure 6 - Scatter plots of evaluation metrics on the test set, including F1-score, precision, recall, and AP50. Each point corresponds to one experimental run. The horizontal axis represents the labelling strategy, while cross-validation folds are indicated by distinct markers, and repeated runs is reflected through color alpha.



Figure 7 - Representative prediction examples from the best-performing configuration – (box_fit, fold1, run1). Bounding boxes indicate predicted insulator instances, with color denoting the predicted class.

5 Discussion

5.1 Result Interpretation and Hypothesis Validation

The results demonstrate that bounding-box supervision provides advantages over image-level labels, even when evaluation is restricted to image-level defect flagging. Models trained with fitted bounding boxes achieve higher accuracy and markedly lower variability across folds and runs, indicating that spatial supervision steers feature learning toward defect-relevant regions.

Training dynamics further explain this behaviour. Fitted-box models converge stably with low run-to-run variability, suggesting optimization toward well-defined minima. In contrast, full-image clamped supervision shows high variability across runs and epochs, with isolated validation peaks that are not sustained, indicative of overfitting to spurious visual cues.

Fold-wise trends offer complementary insights. Folds 1 and 2, corresponding to larger bounding boxes and closer viewpoints, achieve the strongest and most stable performance. Fold 4, dominated by smaller objects, shows degraded results for both strategies, highlighting the challenge of small-scale defect recognition and the added value of annotations spatial detail.

Test-set evaluation further differentiates the approaches. Full-image clamped supervision can achieve high recall for broken insulators but suffers from low precision (~0.5), reflecting biased predictions given the class prevalence (422/792). Fitted-box supervision produces a more balanced precision–recall trade-off. Overall, these results confirm that object detection supervision enhances discriminative representations beyond localization-specific tasks.

5.2 Challenges and Limitations

While the results are encouraging, several limitations should be noted. The study focuses on a single defect type (broken insulators) and assumes mostly one object per image, limiting generalization to multi-defect or dense scenes. The dataset consists of visible-light imagery only, and other sensing modalities (e.g., thermal or multispectral) may yield different results. Only the YOLOv11 Medium architecture was evaluated; other architectures or model scales might interact differently with spatial supervision. Finally, the evaluation was constrained to defect detection, leaving the full impact of localization on downstream tasks unexplored.

5.3 Opportunities and Future Directions

Building on these findings, several avenues for future research are evident. Extending the study to multi-defect, dense scenes would help quantify the advantage of bounding-box supervision in more complex scenarios. Multi-sensor fusion incorporating point clouds [11] for precise measurements, infrared thermal (IRT) for hotspot detection, and ultraviolet imaging (UVI) for corona detection could further improve defect detection [12]. Evaluating different model architectures, including segmentation networks, would clarify the generality of these results across deep learning frameworks. Exploring semi-supervised or weakly-supervised learning could reduce annotation costs while retaining the benefits of spatial supervision. Finally, applying this methodology to other industrial NDT tasks and inspections in complex environments could demonstrate the broader impact of bounding-box guidance.

6 Conclusion

6.1 Summary

This work investigated the role of spatially detailed annotations in training deep learning models for defect inspection of electrical grid assets. Using a controlled experimental setup with YOLOv11 medium, we compared fitted bounding boxes against full-image "clamped" labels while keeping the dataset, backbone, and training hyperparameters constant. Experiments on a real-world dataset of insulators demonstrated that bounding-box annotations improve both accuracy and training stability. Models trained with fitted boxes converge faster and with lower run-to-run variability, suggesting that spatial supervision helps the network focus on defect-relevant regions and suppress irrelevant features. Although region-level annotations introduce modest additional labeling effort, the improved supervision may reduce the number of images required to achieve a given level of performance compared with image-level labels, where many examples are needed to distinguish meaningful features from background noise.

6.2 Takeaway

Spatial supervision enhances feature learning: Bounding boxes provide localized feedback that guides the model toward discriminative regions, improving both precision and recall.

Improved training efficiency: Faster convergence and reduced variability indicate that bounding-box supervision can reduce the effective dataset size required for robust learning.

Limitations of global labels: Image-level supervision alone is more prone to overfitting on spurious cues and shows high variability, particularly in small-object or complex scenarios.

Scale matters: Model generalization is sensitive to object size, emphasizing the importance of including a representative range of scales during training.

6.3 Contributions

Empirical evidence of bounding-box value: Demonstrated that object detection supervision improves feature learning even for simplified, image-level defect flagging.

Controlled comparison framework: Introduced a methodology to isolate the impact of target granularity by using the same dataset, model, and evaluation metrics across supervision types.

Insights on training dynamics: Showed that bounding-box supervision reduces variability and accelerates convergence, indicating more stable optimization.

Practical implications: Highlighted that spatially detailed labels can improve both efficiency and performance in industrial visual inspection, guiding annotation and deployment strategies.

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